

Joint mechanisms

MLS p. 81

Set of 6 mechanisms, *lower pairs*:

- ▶ revolute joint
- ▶ prismatic joint
- ▶ helical joint
- ▶ cylindrical joint
- ▶ spherical joint
- ▶ planar joint

Subgroup of $SE(3)$

Robot manipulators: constructed by connecting lower-pair joints with rigid links — Chapter 2 theory used for analysis

Joint mechanisms

Robot manipulators: constructed by connecting lower-pair joints with rigid links — Chapter 2 theory used for analysis

Chapter 3 & 4 focus on kinematics, dynamics and control of *open-chain* robot manipulators: number of links attached serially by a set of actuated joints

By controlling forces and torques on links: move robot in a specific way

Product of exponentials formula: kinematics of open-chain mechanism as the product of exponentials of twists

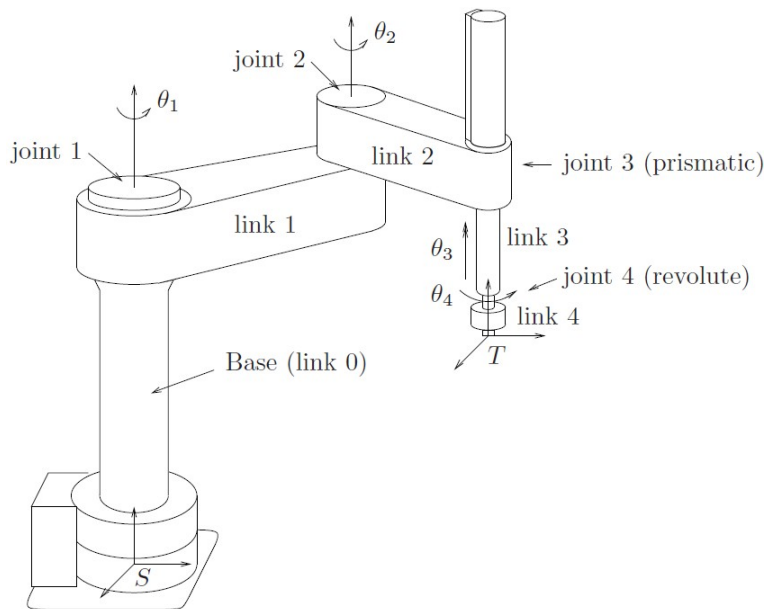
§2 Forward kinematics

Determines the configuration of the end-effector (the gripper or tool mounted on the end of the robot) given the relative configurations of each pair of adjacent links of the robot.

Notation convention:

- ▶ Number joints from 1 to n and links from 0 to n , starting at base (link 0).
- ▶ joint i connects link $i - 1$ with link i .

Notation convention



§2 Forward kinematics

Joint space Q of a manipulator: all possible joint variables (configuration space of robot).

- ▶ Revolute joint: angle $\theta_i \in \mathbb{S}^1$
- ▶ Prismatic joints: linear displacement $\theta_i \in \mathbb{R}$
- ▶ Q is the Cartesian product between each of the individual joint spaces. Fig. 3.1:

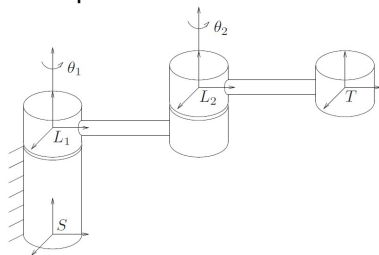
$$\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{R} \times \mathbb{S}^1 = Q$$

Base frame S and tool frame T

§2 Forward kinematics

- ▶ Given $\theta \in Q$
- ▶ To be determined: configuration of T relative to S , represented by mapping $g_{st} : Q \rightarrow SE(3)$

Example:



$$g_{st}(\theta_1, \theta_2) = g_{s l_1}(\theta_1) g_{l_1 l_2}(\theta_2) g_{l_2 t}$$

Mapping $g_{st} : \mathbb{T}^2 \rightarrow SE(3)$ represents forward kinematics of manipulator.

Configuration of T as function of joint angles

§2 Forward kinematics

$g_{l_{i-1}l_i}(\theta_i)$ defines transformation between adjacent link frames, overall kinematics:

$$g_{st}(\theta) = g_{s l_1}(\theta_1) g_{l_1 l_2}(\theta_2) \cdots g_{l_{n-1} l_n}(\theta_n) g_{l_n t} \quad (3.1)$$

Important: the coordinates/displacement components of $g_{l_{i-1}l_i}$ must be expressed in frame $i - 1$.

§2.2 Product of exponentials formula

Rigid motion associated with rotating and translating along axis of twist ξ :

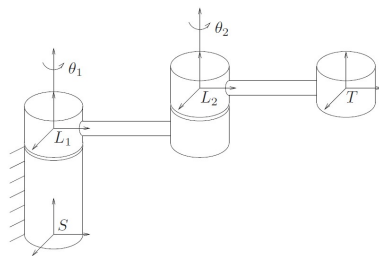
$$g_{ab}(\theta) = e^{\hat{\xi}\theta} g_{ab}(0) \quad (2.37)$$

Fix, at first, joint 1:

$$g_{st}(\theta_2) = e^{\hat{\xi}_2\theta_2} g_{st}(0)$$

Now fix joint 2 and move joint 1 only:

$$\begin{aligned} g_{st}(\theta_1, \theta_2) &= e^{\hat{\xi}_1\theta_1} g_{st}(\theta_2) \\ &= e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} g_{st}(0) \end{aligned} \quad (3.2)$$



ξ_1 & ξ_2 defined on initial configuration, expressed in frame S

θ_2 relative to L_1 , θ_1 to S

§2.2 Product of exponentials formula

Now move joint 1 first: $g_{st}(\theta_1) = e^{\hat{\xi}_1 \theta_1} g_{st}(0)$

This moves the axis of θ_2 ; rotation θ_2 about new axis

$$\xi'_2 = \text{Ad}_g \xi_2 \quad \text{where} \quad g = e^{\hat{\xi}_1 \theta_1}$$

Lemma 2.13 p. 56: twist with coordinates $\text{Ad}_g \xi_2 \in \mathbb{R}^6$ has the corresponding form $g \hat{\xi}_2 g^{-1} \in \mathfrak{se}(3)$.

Used before identity (eq. (2.35) p. 43):

$$e^{\hat{\xi}'_2 \theta_2} = e^{g \hat{\xi}_2 g^{-1} \theta_2} = g e^{\hat{\xi}_2 \theta_2} g^{-1} = e^{\hat{\xi}_1 \theta_1} \left(e^{\hat{\xi}_2 \theta_2} \right) e^{-\hat{\xi}_1 \theta_1}$$

§2.2 Product of exponentials formula

After moving joint 1 first: $g_{st}(\theta_1) = e^{\hat{\xi}_1 \theta_1} g_{st}(0)$

$$e^{\hat{\xi}'_2 \theta_2} = e^{g \hat{\xi}_2 g^{-1} \theta_2} = g e^{\hat{\xi}_2 \theta_2} g^{-1} = e^{\hat{\xi}_1 \theta_1} \left(e^{\hat{\xi}_2 \theta_2} \right) e^{-\hat{\xi}_1 \theta_1}$$

Now fix joint 1 and move joint 2 only:

$$\begin{aligned} g_{st}(\theta_1, \theta_2) &= e^{\hat{\xi}'_2 \theta_2} e^{\hat{\xi}_1 \theta_1} g_{st}(0) \\ &= e^{\hat{\xi}_1 \theta_1} \left(e^{\hat{\xi}_2 \theta_2} \right) e^{-\hat{\xi}_1 \theta_1} e^{\hat{\xi}_1 \theta_1} g_{st}(0) \\ &= e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} g_{st}(0) \end{aligned}$$

§2.2 Product of exponentials formula

Procedure:

- ▶ Define Ref. Config. of manipulator: when $\theta = 0$; $g_{st}(0)$ corresponding transformation between S and T .
- ▶ For each joint i construct ξ_i , screw motion of joint i , with all the other joints fixed.
- ▶ For revolute joint and prismatic joint, respectively:

$$\xi_i = \begin{bmatrix} -\hat{\omega}_i q_i \\ \omega_i \end{bmatrix} \quad \xi_i = \begin{bmatrix} v_i \\ 0 \end{bmatrix}$$

$\omega_i \in \mathbb{R}^3$, $q_i \in \mathbb{R}^3$, $v_i \in \mathbb{R}^3$, all specified relative to S

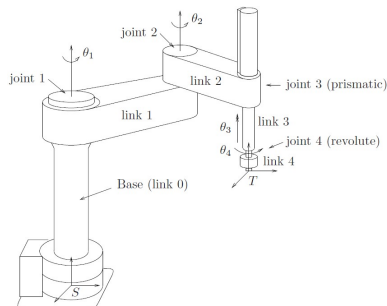
- ▶ $g_{st} : Q \rightarrow SE(3)$ given by

$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0) \quad (3.3)$$

§2.2 Product of exponentials formula

Try to replicate the results of Examples 3.1 and 3.2 symbolically on a computer, using either MATLAB's symbolic toolbox, or Python's sympy module.

§2.3 Parameterization of manipulators via twists



- Choice of base frame

If S lines up with T in the initial configuration, $g_{st}(0) = I$ and

$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_n \theta_n}$$

- Specifying the reference configuration

§2.3 Parameterization of manipulators via twists

Try to replicate the results of Example 3.3 symbolically on a computer, using either MATLAB's symbolic toolbox, or Python's sympy module.

§2.3 Parameterization of manipulators via twists

- ▶ Relationship with the Denavit-Hartenberg (D-H) parameters (p. 93)
 - ▶ Given the D-H parameters for a manipulator: one can determine ξ_i , $\forall i$ by parameterizing g_{l_{i-1}, l_i} using eq. (3-7) p. 94
 - ▶ then use eq. (3-10) to determine ξ_i
 - ▶ Usually easier constructed directly from known geometry.
 - ▶ With product of exponentials: used only two frames S and T (attractive feature).
 - ▶ This and geometric significance of ξ_i -s makes product of exponentials a superior alternative to D-H.